

Obstacle Avoidance Strategy with Path-following for USV Based on Obstacle-guidance Model Using 2D-Lidar in Urban River Conditions

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Abstract—An obstacle detection and automatic avoidance algorithm using lidar based on the obstacle-guidance model is proposed for Unmanned Surface Vehicles (USV) in urban river conditions. The smallest bounding rectangles represent obstacles detected by 2d-lidar due to the clutter of urban river laser data and the shape of actual blocks during navigation. On this basis, an obstacle vortex guidance model is proposed. The obstacle avoidance algorithm takes the heading and speed obtained by the Line-Of-Sight (LOS) method as input. It combines the obstacle guidance model to give the speed and yaw required for collision avoidance, which achieves good real-time performance and robustness in practical applications. Finally, the actual experiment is completed based on the built USV platform, where the entire algorithm is implemented on the Robot Operating System (ROS). The experiment results verify the proposed obstacle-avoidance method's feasibility and validity.

Index Terms—USV, 2d-lidar, obstacle detection, autonomous obstacle avoidance, obstacle-guidance model

I. INTRODUCTION

Many Unmanned Ground Vehicles (UGVs) and Unmanned Aerial Vehicles (UAVs) are appearing in people's vision in recent years. As an essential branch of intelligent robots, Unmanned Surface Vehicles (USVs) are widely deployed for various open water missions like mapping environments, military tasks, and environment monitoring. With the growing demand for water environment management, USVs have been developed for water quality monitoring, water surface garbage cleaning, and other applications in the urban river [1].

To accomplish the surface missions of the USV, collision avoidance is essential and can be divided into two parts: global planning and local obstacle avoidance [2]. For the given global path, the Line-Of-Sight (LOS) method is adopted as the path tracking module in this work, and many studies have been done on the LOS method [3, 4].

The environment perception of USVs is the basis for the fundamental functions of autonomous navigation, especially for

local autonomous collision avoidance. Lidar has the advantages of high ranging accuracy, high angular resolution, and good anti-interference, making it more popular in unmanned driving [5]. More than just detecting obstacles, Ref. [6] uses deep learning for semantic segmentation of three-dimensional point cloud data, identifying bridges, river banks, and vegetation. In this paper, a low-cost 360-degree 2d-lidar is used, which can have good real-time processing performance without a lot of computing power.

The local obstacle avoidance algorithm combined with path-following is the focus of this paper. Classical algorithms include artificial potential field (APF) [7], visible graph [8], and improved algorithms [9, 10]. Different from the radial repulsion field of the APF, a model of the guiding force field along the tangential direction of the obstacle is proposed, through which global path planning, obstacle avoidance, and cruising along the barrier can be performed [11, 12]. However, these methods ignore that the sensor can only detect a part of the shape of the unknown obstacle. In some cases, the obstruction cannot be abstracted into a circle, and the method is challenging to integrate with the actual obstacle data.

The paper focuses on obstacle avoidance in the narrow inland river and proposes a 2D lidar-based obstacle guidance model for urban rivers. The collision avoidance algorithm considers the significant disturbance of the water surface scanning lidar data as well as the actual shape of the object. It uses bounding boxes to represent the obstacles detected by the lidar, each of which has a tangential vortex-like guiding force on the surrounding water. The yaw and speed from the LOS method are combined with the guidance to give the heading and velocity to achieve the purpose of collision avoidance. At the same time, it can return to the predetermined route without re-planning the path. Then, use the Robot Operating System (ROS) to implement each algorithm module.

The remainder of this paper is organized as follows. The LOS, detection and obstacle avoidance algorithms with the guidance model of this paper are detailed in section II. In

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section III, the entire platform and program architecture of this experiment are presented. Section IV mainly shows the experimental results. The conclusion of the paper is given in section V.

II. MAIN RESULTS

A. LOS method

This paper is based on the LOS guidance that commonly used in path-following, is to align the vessel's heading with the LOS angle (LOS angle), and recalled below [3]. The controlled vehicle can reach the desired position for trajectory tracking after a while, and the detailed algorithm model is depicted in Fig. 1.

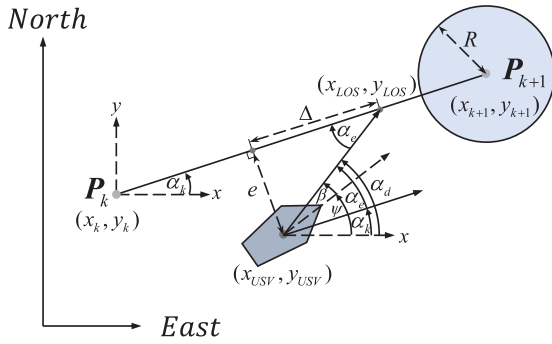


Fig. 1. LOS method model

As shown in Fig. 1, the desired heading α_d in East-North-Up (ENU) coordinates consists of two parts:

$$\alpha_d = \alpha_k + \arctan\left(-\frac{e}{\Delta}\right). \quad (1)$$

where α_k , defined in (2), is the angle between the desired path $P_k P_{k+1}$ and the due east direction, while e is the cross-track error given by (3), and Δ denotes the lookahead distance, which is generally 2 to 5 times the length of the vehicle.

$$\alpha_k = \arctan2(y_{k+1} - y_k, x_{k+1} - x_k). \quad (2)$$

Here cross-track error e is obtained by vector cross product:

$$e = (x_{k+1} - x_{USV}) \sin(\alpha_k) - (y_{k+1} - y_{USV}) \cos(\alpha_k). \quad (3)$$

Then, the USV with the yaw of Ψ needs its heading changed $\beta = \alpha_d - \Psi$, which is also the global angle data for the subsequent obstacle-avoidance algorithm.

During the navigation, the switching condition of the target waypoint is that the position of USV is within the circle with the radius R at point P_{k+1} . Then the next waypoint P_{k+2} is selected as the goal. This acceptable area is defined as:

$$(x_{k+1} - x_{USV})^2 + (y_{k+1} - y_{USV})^2 \leq R^2. \quad (4)$$

where radius R is typically equal to two USV lengths.

B. Obstacle detection

Real-time obstacle detection is the primary goal of USVs' environmental perception, and it is also the premise for surface vehicles to avoid obstacles and plan local paths autonomously. This paper uses the 2d-lidar RPLIDAR-S1 as the only obstacle perception device, while the recognition based on 2d-lidar is divided into three steps: data preprocessing, point cloud clustering, and representation of obstacles.

1) *Data preprocessing*: First, each laser point (r_i, θ_i) is filtered utilizing a $2n + 1$ sliding median filter window to filter out the occasional noise points:

$$r(i) = \text{Mid}(r(i-n), \dots, r(i), \dots, r(i+n)). \quad (5)$$

Then, the scan points are converted from polar coordinates (r_i, θ_i) to Cartesian coordinates (x_i, y_i) , where subsequent clustering and avoidance algorithms are performed. Let the pose of the lidar in the vehicle's body coordinate system is $(t_x, t_y, \theta_{base})$, and the final conversion equation is

$$\begin{bmatrix} x_i \\ y_i \end{bmatrix} = r_i \begin{bmatrix} \cos(\theta_i + \theta_{base}) \\ \sin(\theta_i + \theta_{base}) \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}. \quad (6)$$

2) *Nearest-neighbor clustering*: The nearest-neighbor clustering algorithm, which changes the segmentation threshold according to the distance of the laser point, is selected because the distance between two adjacent points increases with the distance from the lidar. Let $d_r = \max(\alpha r_i \sin(\omega), Width)$ be

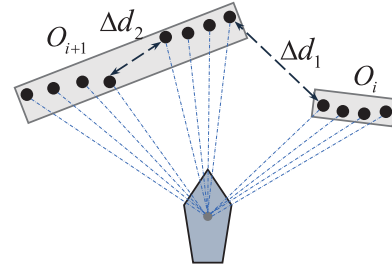


Fig. 2. Segmentation and merging

the threshold of clustering distance, where ω is the lidar angular resolution, while α is the value that the user-specified to control the threshold, and $Width$ means the minimum distance that the surface vehicle can safely pass. If the distance, such as Δd_2 in Fig. 2, between point i and point $i + 1$ is less than the threshold d_r , they are in the same category.

3) *Minimum bounding rectangle*: After clustering, the smallest bounding rectangle is used to represent the obstacle. The rectangle includes the position, length W , width L , and rotation angle λ (the angle between the long side of the rectangle and the x axis) of the obstacle. Here, the rectangle is represented by four vertices. This paper uses the principal component analysis (PCA) method to obtain the minimum circumscribed rectangle. The following are the steps of the method:

- (a) For N laser points $p_i = [x_i, y_i]^T$ ($i = 1, 2, \dots, N$) in the obstacle, find the mean of these points $\bar{p}$:

$$\bar{p} = \frac{1}{N} \sum_{i=1}^N p_i. \quad (7)$$

- (b) Calculate covariance S :

$$S = \frac{1}{N-1} \sum_{i=1}^N (p_i - \bar{p})(p_i - \bar{p})^T. \quad (8)$$

- (c) Singular value decomposition (SVD)

$$S = U \Sigma V^T. \quad (9)$$

- (d) Find the largest eigenvalue from Σ and the corresponding eigenvector in U . Define the eigenvector as v . Then, λ is obtained by:

$$\lambda = \arctan \left(\frac{v(2)}{v(1)} \right). \quad (10)$$

- (e) Finally, four vertices q_i ($i = 1, 2, 3, 4$) need to be determined. Rotate the point cloud by $-\lambda$ angle, shown in Fig. 3. Then, traverse to find the maximum and minimum values on the x-axis and y-axis, which are $[x_{min}, x_{max}]$ and $[y_{min}, y_{max}]$, then arrange and combine them into four vertices q'_i ($i = 1, 2, 3, 4$), and rotate back in the opposite direction by λ angle.

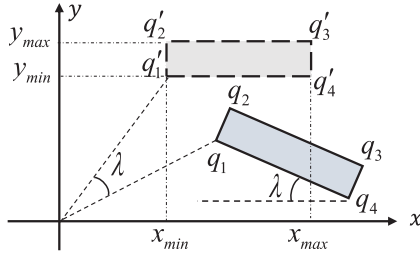


Fig. 3. Determine four vertices of rectangle

C. Obstacle avoidance

In local obstacle avoidance, the commonly used artificial potential field method mainly has two problems: local minimum point and large turning angle. The local minimum problem is shown in Fig. 4. The attractive and repulsive forces cancel out at a point near the obstacle, and the USV will fall to a local minimum. The large turning angle is caused by the sudden change of the resultant force after the USV enters the effective range of the obstacle repulsion. Ref. [9] proposed a method of angle compensation to solve the above problems of APF.

Similar to the idea of giving an offset angle when approaching obstacles, in this article, there is a virtual vortex around each block and a tangential guiding force. Under this assumption, the previous case of local minima can be addressed directly, as shown in Fig. 5. The guidance directions of the obstacles in Fig. 5 are different, which are related to the left or right positions of the blocks on the line connecting the USV and the

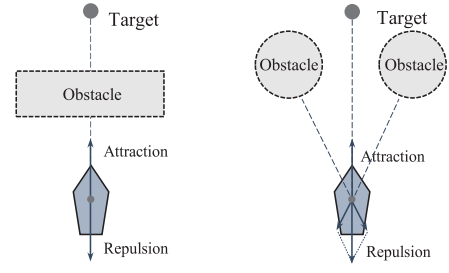


Fig. 4. Local minimum point

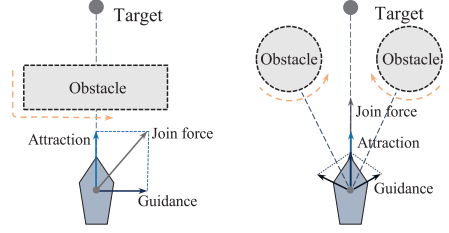


Fig. 5. Obstacle guidance in APF local minimum

target. On the left, the direction is counterclockwise; otherwise, it is clockwise.

Under inland river conditions, the length and width of obstacles detected by lidar cannot be ignored to abstract blocks into circles, and obstacle rectangles represent the obstacles here. Combined with the above analysis, an obstacle guidance model is proposed. As shown in Fig. 6. The influence area

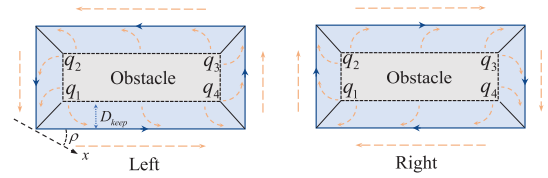


Fig. 6. Obstacle guidance module

of the obstacle is divided into four foundation parts based on the distance between the USV and the four sides of the rectangle. Suppose the edge distance between USV and the nearest obstacle rectangle is D , and the angle between the direction of the edge and the x -axis is ρ . The guidance direction χ of the obstacle to the point where the USV is located is:

$$\chi = \begin{cases} \rho & (D > D_{keep}) \\ \rho + F_p(q_c, \beta) \frac{\pi(D_{keep} - D)}{2D_{keep}} & (D < D_{keep}) \end{cases} \quad (11)$$

where the function $F_p(q_c, \beta)$ is used to judge whether the obstacle is on the left or right of the global path in (12), while β is the global direction obtained in section II-A, and $q_c = \frac{1}{4} \sum_{i=1}^4 q_i$ is the center point of the rectangle. D_{keep} is the artificially set specific distance between the USV and the obstacle to be maintained. If the distance D is smaller than D_{keep} , the obstacle will add repulsion to the vehicle.

$$F_p(q_c, \beta) = \begin{cases} -1 & ((\arctan2(q_c.y, q_c.x) - \beta) > 0) \\ 1 & ((\arctan2(q_c.y, q_c.x) - \beta) \leq 0) \end{cases} \quad (12)$$

After defining the obstacle guidance model, combine the model with the global path. There is a user-defined perception area in front of the USV, which rotates according to the global direction β , as shown in Fig. 7. For each obstacle O_i in the sensing area, the guiding force direction χ_i needs to be calculated by (11).

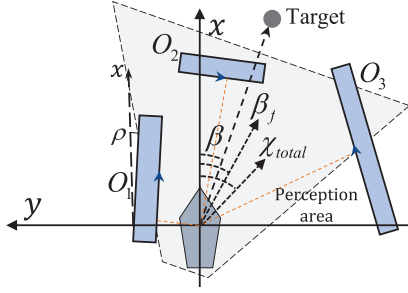


Fig. 7. Schematic diagram of the effect of the obstacle guidance

To combine the forces from the influence of multiple obstacles, the weight W of each barrier needs to be calculated:

$$W = \begin{cases} \frac{1}{D^2} & (F_p(q_c, \beta)(\rho - \beta) \leq 0) \\ 0 & (F_p(q_c, \beta)(\rho - \beta) > 0) \end{cases} \quad (13)$$

The weight W is inversely proportional to the square of the distance D . At the same time, for the obstacle that USV is moving away from, the influence weight is 0. For instance, the weight of O_1 in Fig. 7 is 0. Because the global direction of the surface vehicle is on the left, while the obstacle O_1 is on the right side of the vehicle, and the guidance direction is on the right, which is staggered from the actual driving direction, the USV is away from the obstacle O_1 , and its weight is set to 0. The angle χ_{total} is obtained by the weighted average of the impact of various obstacle:

$$\chi_{total} = \frac{\sum_{i=1}^n \chi_i W_i}{\sum_{i=1}^n W_i} \quad (14)$$

Finally, the guidance of obstacles also needs to be combined with the global direction to obtain the final heading β_f in the body coordinate system:

$$\beta_f = \begin{cases} \beta & (D_{min} > D_{start}) \\ k\chi_{total} + (1-k)\beta & (D_{keep} \leq D_{min} \leq D_{start}) \\ \chi_{total} & (D_{min} < D_{keep}) \end{cases} \quad (15)$$

where D_{min} is the distance to the nearest obstacle in the sensitive area, while D_{start} is the distance set by the user to start autonomous obstacle avoidance, and k is the weight of the algorithm to choose the guidance direction of the obstacle:

$$k = \frac{D_{start} - D_{min}}{D_{start} - D_{keep}} \quad (16)$$

To ensure that the USV do not crack obstacles, add a detection area as shown in Fig. 8, and the forward speed u is required:

$$u = \begin{cases} V_h & (D > D_{start}) \\ \frac{V_h(D - D_{safe}) + V_l(D_{start} - D)}{D_{start} - D_{safe}} & (D_{safe} \leq D \leq D_{start}) \\ V_l & (D < D_{safe}) \end{cases} \quad (17)$$

where V_h is the higher speed when the USV sails, and V_l denotes the lower speed when avoiding obstacles, which even can be negative. D_{safe} is the closer distance than D_{keep} to make sure the safe navigation of the vehicle.

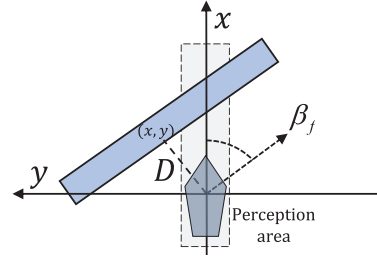


Fig. 8. Schematic diagram of obstacle avoidance algorithm

III. SYSTEM OVERVIEW

The USV platform used in this paper is a catamaran with two thrusters, as shown in Fig. 9. It uses differential speed to realize the function of steering. Such a design can reduce the rolling motion caused by the low mass of the vehicle. The following is a detailed introduction to the surface vehicle.



Fig. 9. USV in field experiment

A. Hardware platform

The hardware platform of the USV is the basis of its control, including MCU, control computer, motors, and various sensors, as shown in Fig. 10. The power supply module comprises two lithium batteries and various voltage stabilizing modules to power the other modules. The STM32f407 is selected for the MCU to collect different sensor data and control motors depending on the instructions of the control computer. The main functions of the control computer are communicating with the MCU, receiving lidar data, communicating with the upper computer, and processing each sensor data or the host

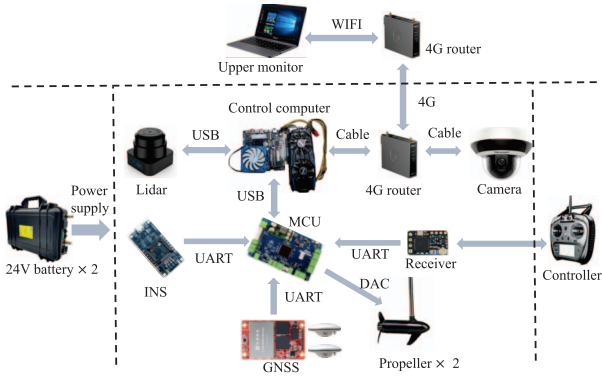


Fig. 10. Hardware system of USV

computer instructions. Sensors include dual-antenna GPS, IMU, lidar, and camera.

B. Software platform

The software platform of USV is the foundation for its realization of various functions. The software platform on USV is divided into two parts: the microcontroller's control algorithm and the control computer's processing part.

1) *MCU*: To be compatible with remote controller and control computer commands, the MCU measures the actual angular velocity through the IMU and then uses the PID control algorithm to perform closed-loop control of the rotation velocity. The closed loop of the heading is carried out on the computer.

2) *Robot Operating System*: Use ROS as the software architecture for writing robot programs in the control computer. The communication of each node function can be realized very simply by publishing and subscribing to messages. Fig. 11

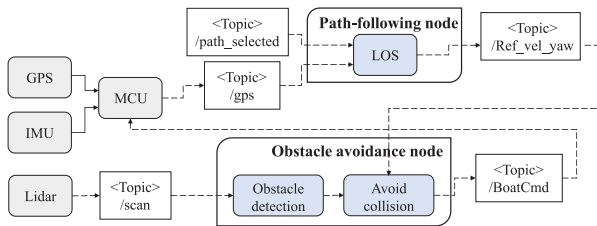


Fig. 11. The nodes and messages run in ROS

shows the primary nodes and messages run in ROS. The MCU sends a series of sensor data to ROS through the *rosserial* package and receives the control commands of the computer. The path-following node subscribes to the given path and current GPS positioning data and provides the reference with speed and heading to the obstacle avoidance node through the LOS method. The obstacle avoidance node combines the scanning data of lidar for obstacle identification and collision avoidance and finally sends speed and angular velocity commands to the MCU.

IV. EXPERIMENTS AND RESULTS

The automatic collision avoidance experiment of the USV during the path following is carried out in an urban river of Hangzhou. The experimental platform is shown in Section III-B, while the obstacle is also the self-developed and manufactured USV shown in Fig. 12. The host computer sends the target waypoints to the USV platform in the experiment. As shown in Fig. 13, the USV implements path tracking with the LOS method and performs real-time obstacle avoidance by combining the laser data from the 2d-lidar on the way.



Fig. 12. The actual scene of obstacle avoidance

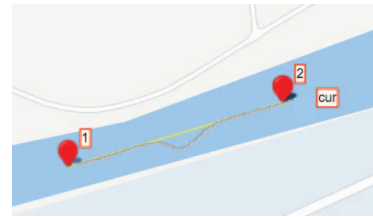


Fig. 13. Display on the host computer

As shown in Fig. 14, it is confirmed that the obstacle avoidance algorithm proposed in the article has achieved satisfactory results in practical experiments, where the green dotted line is the predetermined trajectory, while the red dotted line box is the obstacle. while the red dotted line box is the

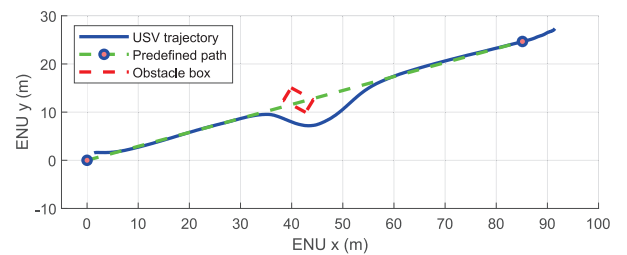


Fig. 14. Automatic obstacle avoidance during path following. The manually selected path is marked with a green dotted line, the actual track of the USV is blue, and the red is the position of the obstacle USV.

obstacle. It can be seen that the blue line, which is the actual route of the USV, bypasses the obstacle well and returns to the predetermined green trajectory. When the lidar detects an obstacle, the local collision avoidance algorithm combined with the global direction can make the USV avoid the obstacle and

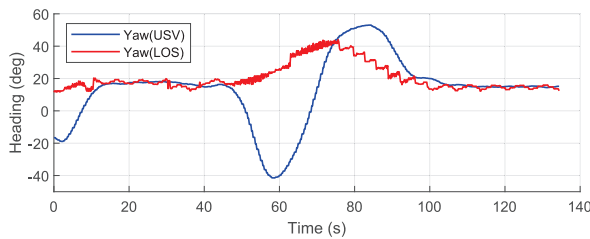


Fig. 15. USV heading Measured and reference heading given by LOS method.

return to the global path when there is no obstacle. Fig. 15 compares the heading given by the LOS method with the actual yaw of the vehicle. It can be concluded that the heading angle of the USV is consistent with the reference when no obstacle is found in front of the vehicle. In contrast, when avoiding obstacles, the directions of the two are separated and tend to be the same after obstacle avoidance.

V. CONCLUSION

This paper studies the obstacle avoidance problem using 2d-lidar in urban river conditions. A guiding force model of obstacles to the surrounding waters is proposed based on the direction of the global path, obtained through the LOS method, and the information of the obstacle boxes after clustering the lidar data. Based on the obstacle guidance model, the algorithm to get the USV target heading angle and velocity is given. The USV can avoid obstacles and return to the route without replanning the path by combining the directions of multiple obstacle guidance forces and the global approach. The experimental results confirm that the obstacle avoidance model and algorithm can meet the collision avoidance requirements of USV in urban rivers.

The future work will add the kinematic model of the unmanned vehicle to the obstacle avoidance algorithm and adjust the corresponding obstacle avoidance parameters according to the performance of the USV. In addition, combine with the LiDAR attitude during navigation to calibrate the data. Consider high-speed obstacle avoidance and the impact of dynamic obstacles in the long-term research.

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